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Compact E-axle assembly

Abstract

An e-axle assembly includes a housing extending from a first housing end to a second housing end to define an internal compartment. At least one electric motor is disposed in the internal compartment and includes a stator and a rotor rotatable relative to the stator about an axis. The rotor defines a hollow extending internally within the rotor, and at least one of a gearing mechanism, a shifting actuator, and/or a differential is disposed within the hollow (and thus packaged within the rotor). In a preferred arrangement, all and any of the gearing mechanisms, shifting actuators, and differential are disposed within the hollow of the rotor. This internal packaging of the drive components within the rotor reduces an overall space and weight of the e-axle assembly relative to the prior art designs to more easily and readily achieve the smaller space and weight allotments afforded by electric and/or hybrid vehicles.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) The subject application claims priority to U.S. Provisional Patent Application Ser. No. 63/388,267 filed on Jul. 12, 2022, the entire disclosure of which is incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to live axles for vehicles, including both solid beam-type axles and independently suspended axles. More particularly, the present invention relates to an e-axle assembly for an electric or hybrid vehicle.

2. Related Art

(2) This section provides background information related to the present disclosure which is not necessarily prior art.

(3) An e-axle assembly is a powertrain electric drive unit for electric or hybrid vehicles which combines the motor, inverter, power electronics and other transmission elements together, aiming to make electric drive of the related electric or hybrid vehicle less complex. The e-axle assembly seeks a compact and cost effective design as a result of integrating the electric motor, power electronics and gearbox in one unit.

(4) However, traditional e-axle assemblies require large package space claims to allow for the large motor and gear reduction structures to be integrated into the e-axle assembly. Put another way, the conventional motor structure and parallel axis gearing in traditional e-axle assemblies leads to much larger space claims, as well as higher weight for the center section of the e-axle assembly as compared to conventional live axles. Furthermore, the conventional rotor structures, attached parallel axis or coaxial gearing, and side-gear differential still make the e-axle assembly difficult to fit in the space and weight allotments afforded by the electric or hybrid vehicle.

(5) Accordingly, there is a continuing desire to further develop more compact e-axle assemblies to further reduce their size and weight requirements to facilitate simpler implementation into the space and weight allotments afforded by the electric or hybrid vehicle.

SUMMARY OF THE INVENTION

(6) According to an aspect of the disclosure, an e-axle assembly includes a housing extending from a first housing end to a second housing end to define an internal compartment. At least one electric motor is disposed in the internal compartment and includes a stator and a rotor rotatable relative to the stator about an axis. The rotor defines a hollow extending internally within the rotor and along the axis, and at least one of a gearing mechanism, a shifting actuator, and/or a differential is disposed within the hollow (and thus packaged within the rotor). In a preferred arrangement, all and any of the gearing mechanisms, shifting actuators, and differential are disposed within the hollow of the rotor. This internal packaging of the drive components within the rotor reduces an overall space and weight of the e-axle assembly relative to the prior art e-axle assembly designs to more easily and readily achieve the smaller space and weight allotments afforded by electric and/or hybrid vehicles.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Other aspects and advantages of the present disclosure will be readily appreciated, as the same becomes better understood by reference to the following detailed description when considered in connection with the accompanying drawings wherein:

- (2) FIG. 1 illustrates a first embodiment of an e-axle assembly;
- (3) FIG. 2 illustrates a second embodiment of the e-axle assembly;
- (4) FIG. 3 illustrates a third embodiment of the e-axle assembly;
- (5) FIG. 4 illustrates a fourth embodiment of the e-axle assembly;
- (6) FIG. 5 illustrates a fifth embodiment of the e-axle assembly;
- (7) FIG. 6 illustrates a sixth embodiment of the e-axle assembly;
- (8) FIG. 7 illustrates a seventh embodiment of the e-axle assembly illustrating the additional incorporation of a final drive reduction gearing; and
- (9) FIG. 8 illustrates an eighth embodiment of the e-axle assembly exemplarily illustrating that any of the aforementioned embodiments of the e-axle assembly can be positioned off of the vehicle's center.

DETAILED DESCRIPTION OF THE ENABLING EMBODIMENTS

(10) Example embodiments will now be described more fully with reference to the accompanying drawings. In general, the subject embodiments are directed to an e-axle assembly **10** for an electric or hybrid vehicle. However, the example embodiments are only provided so that this disclosure will be thorough, and will fully convey the scope to those who are skilled in the art. Numerous specific details are set forth such as examples of specific components, devices, and methods, to provide a thorough understanding of the embodiments of the present disclosure. It will be apparent to those skilled in the art that specific details need not be employed, that example embodiments may be embodied in many different forms and that neither should be construed to limit the scope of the disclosure. In some example embodiments, well-known processes, well-known device structures, and well-known technologies are not described in detail.

(11) Referring to the Figures, the e-axle assembly **10** includes a housing **11** (such as a central drive unit housing) which extends from a first housing end **12** to a second housing end **13** to define an internal compartment **14**. At least one electric motor **16** is disposed in the internal compartment **14** and includes a stator **18** and a rotor **20** rotatable relative to the stator **18** about an axis A. As illustrated throughout the Figures, a plurality of surface mount magnets **22** can be disposed along an exterior rotor surface **21** of the rotor **20** for facilitating rotation of the rotor **20** relative to the stator **18**. The rotor **20** defines a hollow **24** extending internally within the rotor **20** and along the axis A, and at least one of a gearing mechanism **26**, a shifting actuator **28**, and/or a differential **30** is disposed within the hollow **24** (and thus packaged within the rotor **20**) to achieve a reduced overall size and weight of the e-axle assembly **10** relative to the prior art e-axle assembly designs.

(12) As best illustrated in FIGS. 1-5, in accordance with an aspect of the disclosure, at least one gearing mechanism **26** and at least one shifting actuator **28** are both disposed within the hollow **24** of the rotor **20**. As further illustrated in FIGS. 1-2, the differential **30** can also be disposed within the hollow **24** (in addition to the at least one gearing mechanism **26** and the at least one shifting actuator **28**), particularly when only one electric motor **16** is disposed within the internal compartment **14** of the housing **11**. In any arrangement, a pair of output shafts **32** extend outwardly from respective first and second housing ends **12**, **13** of the housing **11** for operably coupling the e-axle assembly **10** to the wheels **36** of the electric or hybrid vehicle, either directly or indirectly through wheel-end reducers **38**.

(13) As illustrated in FIG. 1, in accordance with a first embodiment, the e-axle assembly **10** includes a single electric motor **16** which houses a pair of shift packages (i.e., two gearing

mechanisms **26'**, **26''** and two shift actuators **28'**, **28''** each respectively paired with one another) and a differential **30** within the hollow **24** defined by the rotor **20**. In this arrangement, the rotor **20** is operably connected to and drives the differential **30** arranged centrally within the internal compartment **14** as a high speed differential. The differential **30** presents a pair of opposing first and second differential sides **33**, **34**, with the first differential side **32** facing the first housing end **12** and the second differential side **34** facing the second housing end **13**. The at least one gearing mechanism **26** includes a first gearing mechanism **26'** disposed within the hollow **24** adjacent the first housing end **12** and a second gearing mechanism **26''** disposed within the hollow **24** adjacent the second housing end **13**. The first gearing mechanism **26'** is operably connected to the first differential side **33** of the differential **30** via a first gear shaft **40'** and the second gearing mechanism **26''** is operably connected to the second differential side **34** of the differential **30** via a second gear shaft **40''** for driving torque from the differential **30** to the respective first and second gearing mechanisms **26'**, **26''**. The first and second gearing mechanisms **26'**, **26''** are each preferably planetary gear stages fitted with a two-stage shift mechanism which selects output from the planetary sun gear for high range or selects output from the planetary carrier for low range.

(14) The at least one shifting actuator **28** includes a first shifting actuator **28'** and a second shifting actuator **28''** each preferably arranged as two-speed actuators and each operably connected to a respective one of the first and second gearing mechanisms **26'**, **26''** for operating the respective two-stage shift mechanism. A pair of output shafts **32** are each operably connected to a respective one of the first and second gearing mechanisms **26'**, **26''** and extend outwardly from opposing first and second housing ends **12**, **13** of the housing **11** for sending output torque to the wheels **36** or wheel-end reducers **38** of the electric or hybrid vehicle. With reference to FIG. **1** and the aforementioned description, a power flow of the first embodiment of the e-axle assembly **10** proceeds by the rotor **20** driving the differential **30**, the differential **30** driving the first stage suns of the planetary gear stages in each of the first and second gearing mechanisms **26'**, **26''** and the first stage carriers of the planetary gear stages in each of the first and second gearing mechanisms **26'**, **26''** respectively driving one of the output shafts **32**. An optional locker and optional limited slip could be incorporated into the first embodiment without departing from the scope of the subject disclosure.

(15) As illustrated in FIG. **2**, in accordance with a second embodiment, the e-axle assembly **10** similarly includes a single electric motor **16**, but in this instance houses a single shift package (i.e., a single gearing mechanism **26** and a single shifting actuator **28** paired with one another) and a differential **30** within the hollow **24** defined by the rotor **20**. In this arrangement, the rotor **20** is operably connected to and drives the single gearing mechanism **26** arranged as a central planetary reducer. Although the operable connection between the rotor **20** and the single gearing mechanism **26** is illustrated as being arranged in the central portion of the rotor **20**, the operable connection between the rotor **20** and the gearing mechanism **26** can also be offset and adjacent a rotor end of the rotor **20** without departing from the scope of the subject disclosure. The planetary reducer of the single gearing mechanism **26** can be fitted with a two-speed shift mechanism which selects output from the planetary sun gear for high range or selects output from the planetary carrier for low range.

(16) The single shifting actuator **28** is operably connected to the single gearing mechanism **26** for operating the two-speed shift mechanism of the gearing mechanism **26**. An output of the single gearing mechanism **26** is coupled to and drives the differential **30** and a pair of output shafts **32** are operably coupled to the differential **30** on respective first and second differential sides **33**, **34** and extend outwardly from opposing first and second housing ends **12**, **13** for sending output torque to the wheels **36** or wheel-end reducers **38** of the electric or hybrid vehicle. With reference to FIG. **2** and the aforementioned description, a power flow of the second embodiment of the e-axle assembly **10** proceeds by the rotor **20** driving the first stage sun of the planetary reducer in the gearing mechanism **26**, the first stage carrier of the planetary reducer in the gearing mechanism **26**

driving the differential **30**, and the differential **30** driving the pair of output shafts **32**. An optional locker and optional limited slip could also be incorporated into the second embodiment without departing from the scope of the subject disclosure.

(17) In either of the first or second embodiments of the e-axle assembly **10**, the differential **30** can be arranged as a bevel gear or planetary, open or torque transferring (viscous, hydraulic, Torsen, Quaife, dog clutch, or Belleville/conical spring) without departing from the scope of the subject disclosure.

(18) As illustrated in FIGS. **3-5**, in accordance with a third, fourth and fifth embodiment, the e-axle assembly includes a first and second electric motor **16'**, **16''** disposed within the internal compartment **14** of the housing **11**, each adjacent a respective first or second housing end **12**, **13** and in axially spaced relationship with one another. In accordance with the previous disclosure, the first electric motor **16'** includes a first stator **18'** and a first rotor **20'** rotatable relative to the first stator **18'** about the A, and the second electric motor **16''** includes a second stator **18''** and a second rotor **20''** rotatable relative to the second stator **18''** about the axis A. Each of the first and second rotors **20'**, **20''** define respective first and second hollows **24'**, **24''** extending within the rotors **20'**, **20''** and along the axis A. A first gearing mechanism **26'** and a first shifting actuator **28'** are disposed within the first hollow **24'** of the first rotor **20'** and a second gearing mechanism **26''** and a second shifting actuator **28''** are disposed within the second hollow **24''** of the second rotor **20''**. Each of the first and second electric motors **16'**, **16''** are coupled with a respective one of the output shafts **32**, and thus a differential is not included in these embodiments for outputting torque to the wheels **36** or wheel-end reducers **38** of the electric or hybrid vehicle. More specifically, a first output shaft of the pair of output shafts **32** is operably connected to the first electric motor **16'** and extends outwardly from the first housing end **12** for connection to a first wheel of the electric or hybrid vehicle and a second output shaft of the pair of output shafts **32** is operably connected to the second electric motor and extends outwardly from the second housing end **13** for connection to a second wheel of the electric or hybrid vehicle.

(19) The first rotor **20'** is operably connected to and drives the first gearing mechanism **26'** and the second rotor **20''** is operably connected to and drives the second gearing mechanism **26''**. Each of the first and second gearing mechanisms **26'**, **26''** are preferably arranged as planetary reducers, such that the rotors **20'**, **20''** drive the planetary reducers in the first and second gearing mechanisms **26'**, **26''** through a rotor to sun gear connecting hub **42**. Similar to the other embodiments, the planetary reducers of the first and second gearing mechanisms **26'**, **26''** can be fitted with a two-speed shift mechanism which selects output from the planetary sun gear for high range or selects output from the planetary carrier for low range.

(20) The first shifting actuator **28'** is operably connected to the first gearing mechanism **26'** and the second shifting actuator **28''** is operably connected to the second gearing mechanism **26''** for operating the two-speed shift mechanisms. A pair of output shafts **32** are operably connected to a respective one of the first and second gearing mechanisms **26'**, **26''** and extend outwardly from opposing first and second housing ends **12**, **13** for sending output torque to wheels **36** or wheel-end reducers **38** of the electric or hybrid vehicle. With reference to FIGS. **3-5** and the aforementioned description, a power flow of these three embodiments of the e-axle assembly **10** proceeds by each rotor **20'**, **20''** of the first and second electric motors **16'**, **16''** driving first stage suns of the planetary reducers in the respective first and second gearing mechanisms **26'**, **26''** and first stage carriers of the planetary reducers in the respective first and second gearing mechanisms **26'**, **26''** driving a respective one of the output shafts **32**.

(21) In the third embodiment illustrated in FIG. **3**, differentiation occurs via no connection between the wheels **36**. However, as illustrated in FIGS. **4-5**, the fourth and fifth embodiments of the e-axle assembly **10** additionally includes a torque coupling element **44** extending between and operably coupled to the first and second gearing mechanisms **26'**, **26''** housed respectively in the hollows **24'**, **24''** of the rotors **20'**, **20''** for coupling the two, spaced but adjacent first and second electric motors

16', 16''. Put another way, in the fourth and fifth embodiments, the coupling element **44** is added to couple the planetary reducers input (sun gear) or output (planet carrier) of the first and second gearing mechanisms **26', 26''**. This torque coupling element **44** can be arranged as viscous hydraulic, one-way clutches, or dog clutches, as examples. However, other means of torque coupling elements **44** could be utilized without departing from the scope of the subject disclosure. (22) As further illustrated in FIGS. 4-5, in the fourth and fifth embodiments, an electric coupling element **46** can also be added to extend between and couple current supplied to the first and second stators **18', 18''** in each of the first and second electric motors **16', 16''** disposed within the internal compartment **14** of the housing **11**. More specifically, the electric coupling element **46** is operably connected with both of the first and second stators **18', 18''** to establish a physical (contactor or transistor) series or parallel connection of the phase windings in the stators **18', 18''**. The electric coupling element **46** may also be accomplished with coordinated control of the phase current to the first and second electric motors **16', 16''**, with the goal of controlling torque and/or speed of each motor **16**.

(23) The torque coupling element **44** or the electric coupling elements **46** may be used at the same time when full axle torque is required at a single wheel. For example, in this case, the torque coupling element **44** may lock left to right and allow both of the first and second electric motors **16', 16''** to apply torque to a single wheel of the electric or hybrid vehicle. When lower torque is required and higher efficiency is required, the torque coupling element **44** may lock or provide a small amount of left to right speed differentiation and allow a single motor to supply the required torque. In this case, the motor on the side with the higher speed requirement would typically drive, which allows a small amount of slip in the torque coupling element **44** to supply correct speed and torque to the other wheel. This shifting of torque drive from one motor to the other is accomplished with coordinated control of the phase current on both motors **16', 16''** via the electric coupling element **46**, which includes an electric control therebetween to allow the first and second motors **16', 16''** to be controlled as one. In the fourth and fifth embodiments, differentiation occurs via coupling in free running mode, with torque transfer between the wheels facilitated via the torque coupling element **44** and the electric coupling elements **46**.

(24) As illustrated in FIG. 5, the fifth embodiment of the e-axle assembly **10** differs from the fourth embodiment in that the torque coupling element **44** may allow a non-linear angular displacement of the output shafts **32**. In other words, the torque coupling element **44** can allow a non-coaxial and angular connection between the first and second gearing mechanisms **26', 26''** housed in the first and second electric motors **16', 16''**, such that each of the first and second rotors **20', 20''** extend along respective axis A.sub.1, A.sub.2 that are non-parallel and angled relative to one another. For example, this could be accomplished by incorporating a constant velocity joint or non-parallel axis gears into the torque coupling element **44**. The angular displacement between the output shafts **32** may be fixed or variable depending on the needs of the vehicle's suspension.

(25) As illustrated in FIG. 6, in a sixth embodiment, the e-axle assembly **10** can be utilized to drive a front axle **48** connected to a pair of front wheels **49** and a rear axle **50** connected to a pair of rear wheels **51** of the electric vehicle in a 4-wheel drive configuration. In this embodiment, the e-axle assembly **10** is arranged similar to the fourth embodiment (or any other embodiment, for that matter), except that the pair of output shafts **32** extend out of the first and second housing ends **12, 13** and are respectively connected to a front differential **52** of the first axle **48** and a rear differential **54** of the rear axle **50**. Thus, the torque coupling element **44** is arranged to allow locking of the planetary reducers in the first and second gearing mechanisms **26', 26''** to supply full torque to either the front differential **52** or the rear differential **54**. The axle differentials may allow torque coupling from left to right or locking from left to right. Thus, as illustrated in FIG. 6, the sixth embodiment of the e-axle assembly **10** includes the first and second electric motors **16', 16''** each having a respective shift package which can be coupled together via the torque coupling element **44** to drive the front and rear axles **48, 50** of the electric or hybrid vehicle.

(26) Each of the embodiments discussed previously are arranged with the final drive reduction (e.g., wheel end reducers **38**) being located at the wheel-end. However, as illustrated in FIG. 7, each of these embodiments can additionally be modified to locate and incorporate final drive reduction gearing **56** into the e-axle assembly **10**, and thus within the internal compartment **14** of the housing **12**. Put another way, the final drive reduction gearing **56'**, **56''** may be located in the internal compartment **14** and in operable connection with and between the gearing mechanism(s) **26** and the output shafts **32** for each of the respective motors **16'**, **16''**, as shown in the e-axle assembly **10** of FIG. 7. In this case, the final drive reduction gearing **56** may be coaxial/planetary gears, parallel axis gears or non-parallel axis gears.

(27) Each of the embodiments discussed previously are also preferably arranged with the e-axle assembly **10** positioned along the vehicle's center. However, for each of these embodiments, the drive unit may also be positioned off of the vehicle's center, either to the left or the right (as illustrated in FIG. 8) to provide greater package flexibility and make space for other vehicle systems. For example, on a left-hand drive vehicle, the e-drive system may be packaged slightly to the right to allow room for a steering shaft and steering gear unit.

(28) As with be appreciated in view of the foregoing description of the embodiments, the e-axle assembly **10** can provide a live axle motor and first stage reduction in the center, final drive at wheel end or final drive in center, or an independent suspension drive unit motor and first stage reduction in the center, final drive at wheel end or final drive in center. Further, although not described in detail, additional features for consideration include controls unique to live axle or center drive unit, axle beam structures unique to the motor/gearing subsystem, suspension structures unique to the motor/gearing subsystem, and brake/wheel-end/wheel-bearing structures unique to the motor/gearing subsystem. Furthermore, the term live axle is intended to refer to and include both solid beam-type axles and independently suspended axles, and thus the principles disclosed herein are applicable to both types of axles.

(29) It should be appreciated that the foregoing description of the embodiments has been provided for purposes of illustration. In other words, the subject disclosure it is not intended to be exhaustive or to limit the disclosure. Individual elements or features of a particular embodiment are generally not limited to that particular embodiment, but, where applicable, are interchangeable and can be used in a selected embodiment, even if not specifically shown or described. The same may also be varies in many ways. Such variations are not to be regarded as a departure from the disclosure, and all such modifications are intended to be included within the scope of disclosure.

Claims

1. An e-axle assembly for an electric or hybrid vehicle, comprising: a housing extending from a first housing end to a second housing end to define an internal compartment; a single electric motor disposed within said internal compartment and including a stator and a rotor rotatable relative to said stator about an axis; said rotor extending between a first rotor end and a second end and defining a hollow extending internally and continuously within said rotor between said first and second rotor ends; a first gearing mechanism and a first shifting actuator operably paired with one another and disposed within said hollow adjacent said first housing end and a second gearing mechanism and a second shifting mechanism operably paired with one another and disposed within said hollow adjacent said second housing end; said rotor operably connected to and driving a differential arranged centrally within said hollow to present a first differential side facing said first housing end and a second differential side facing said second housing end, wherein said first gearing mechanism is operably connected to said first differential side of said differential via a first gear shaft and said second gearing mechanism is operably connected to said second differential side of said differential via a second gear shaft; and a pair of output shafts each operably connected to a respective one of said first and second gearing mechanisms and extending outwardly from

respective first and second housing ends of said housing for sending torque to wheels of the electric or hybrid vehicle.

2. An e-axle assembly for an electric or hybrid vehicle, comprising: a housing extending from a first housing end to a second housing end to define an internal compartment; a first electric motor and a second electric motor each disposed within said internal compartment in axially spaced relationship with one another and each adjacent a respective first and second housing end; said first electric motor including a first stator and a first rotor defining a first hollow extending within said first rotor; said second electric motor including a second stator and a second rotor defining a second hollow extending within said second rotor; a first gearing mechanism and a first shifting actuator disposed within said first hollow and a second gearing mechanism and a second shifting actuator disposed within said second hollow; said first rotor operably connected to and driving said first gearing mechanism and said second rotor operably connected to and driving said second gearing mechanism, a pair of output shafts operably connected to a respective one of said first and second gearing mechanisms and extending outwardly from respective first and second housing ends for sending output torque to wheels of the electric or hybrid vehicle; and a torque coupling element extending between and operably coupled to said first and second gearing mechanisms housed respectively in each of the first and second hollows, said torque coupling element selectively switchable from an unlocked condition wherein said first and second electric motors are unlocked from one another and individually and separately drive respective wheels of the electric or hybrid vehicle to a locked condition wherein said first and second electric motors are locked to one another to allow one or both of said first and second electric motors to supply torque to a single wheel of the electric or hybrid vehicle.

3. The e-axle assembly as set forth in claim 2, further comprising an electric coupling element extending between and operably coupled with said first and second stators to establish an electric connection between said first and second electric motors.

4. The e-axle assembly as set forth in claim 2, further comprising: said first rotor and said first hollow of said first electric motor arranged along an axis A.sub.1; said second rotor and said second hollow of said second electric arranged along an axis A.sub.2; said axis A.sub.1, A.sub.2 disposed in non-parallel and angled relationship to one another; and said torque coupling element configured to allow a non-angular displacement of said pair of output shafts.
